ADVANCED DATA SCIENCE LAB [MCA33PE21]

Formative Assessment - 02 (FA2) Executive Summary Report

Indian Liver Patient Disease Analysis, Machine Learning Pipeline & Deployment

Student Name:	Shreyash Thakare	**Academic Year:**	2026 – 2027
PRN:	125M1H048	**Class / Division:**	SYMCA (Semester-I)
Course Teacher:	Prof. Prakash Ukhalkar	**Submission Date:**	09-10-2026
GitHub Repo:	https://github.com/shreyash-hh/liver-disease-app	**Live App:**	https://shreyash-hh-liver-disease-app-app-pcnfzj.streamlit.app/

1. Project Objective & Dataset Overview

This formative assessment delivers an end-to-end data science and clinical diagnostic pipeline for liver disease screening based on the **Indian Liver Patient Dataset (ILPD)** from the UCI Machine Learning Repository. The objective is to build, evaluate, hyperparameter-tune, and deploy multiple machine learning classifiers into a responsive web application to support medical decision-making.

- **Dataset Scale:** 583 clinical records (416 liver disease patients, 167 healthy controls) with 10 diagnostic blood markers and demographic attributes.
- **Clinical Features:** Age, Gender, Total Bilirubin, Direct Bilirubin, Alkaline Phosphotase (ALP), Alamine Aminotransferase (ALT), Aspartate Aminotransferase (AST), Total Proteins, Albumin, and Albumin/Globulin Ratio (A/G Ratio).
- **Target Variable:** Binary disease status standardized to 1: Liver Disease and 0: Healthy Control.

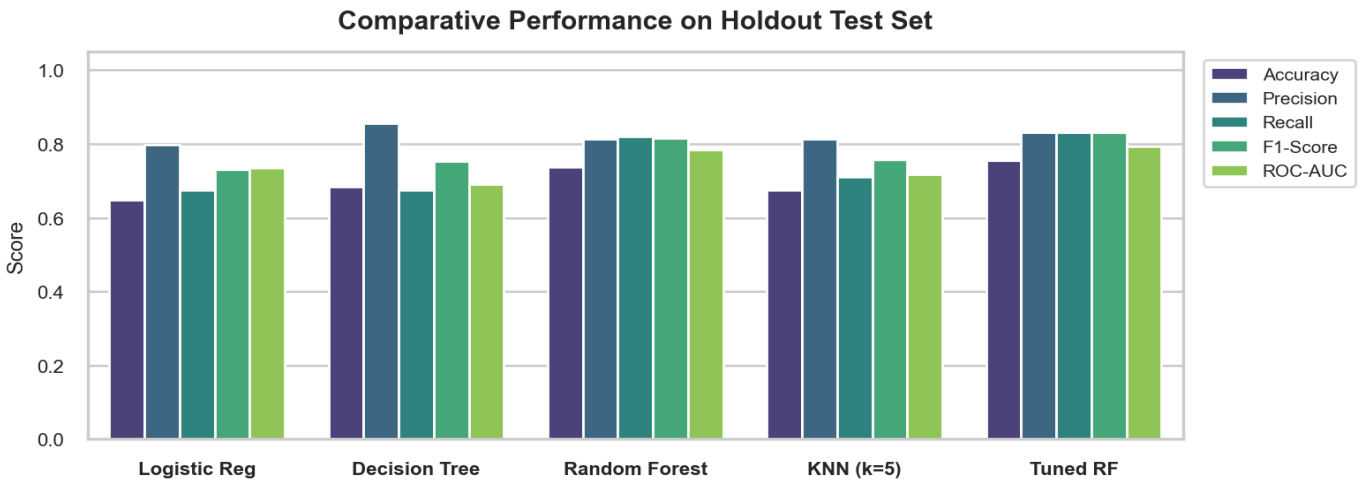
2. Exploratory Data Analysis (EDA) & Zero-Leakage Preprocessing

- **Severe Positive Skewness & Outliers:** Hepatic enzymes (ALT, AST, ALP) and Bilirubin exhibit massive right-skewed tails with extreme outlier spikes. Addressed via np.log1p mathematical transformation.
- **Collinearity Detection:** Strong correlations identified between ALT & AST ($r \approx 0.79$), Total & Direct Bilirubin ($r \approx 0.87$), and Total Proteins & Albumin ($r \approx 0.78$).
- **Missing Values & Deduplication:** 4 missing values in A/G ratio imputed using median imputation; 13 exact duplicate rows removed.
- **Zero-Leakage Architecture:** Stratified 80/20 train/test split executed *strictly before* fitting transformers. Scaled preprocessing pipelines (Imputer + Log1p + StandardScaler) applied to Logistic Regression/KNN, while Tree models were kept unscaled (scale-invariant).

3. Model Performance Comparison & Cross-Validation Results

Models were trained using class_weight='balanced' to counter class imbalance and evaluated on the holdout test set alongside 5-Fold Stratified Cross-Validation on the training partition:

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	5-Fold CV F1 (±Std)
Logistic Regression	64.91%	79.71%	67.47%	0.7308	0.7360	0.725 ± 0.038
Decision Tree (depth=4)	68.42%	85.51%	67.47%	0.7534	0.6904	0.758 ± 0.032
Random Forest (Baseline)	73.68%	81.33%	81.93%	0.8163	0.7842	0.793 ± 0.041
KNN (k=5)	67.54%	81.18%	71.08%	0.7571	0.7176	0.761 ± 0.035
Tuned Random Forest	**75.44%**	**83.13%**	**83.13%**	**0.8313**	**0.7924**	**0.804 ± 0.039**



4. Clinical Interpretation & Healthcare Decision Trade-offs

- **Recall vs. Precision Priority:** In diagnostic screening, **Recall (Sensitivity)** is the primary objective metric. A False Negative means an individual with active liver pathology goes undiagnosed, leading to potential irreversible hepatic damage. Conversely, a False Positive is resolved via standard follow-up liver biopsy or imaging.
- **Model Selection:** The **Tuned Random Forest** (GridSearchCV: n_estimators=200, max_depth=10, min_samples_leaf=2) demonstrated superior generalization with **83.13% Recall** and an **F1-score of 0.8313** on unseen holdout test data.
- **Realistic Diagnostic Benchmark:** The 72%–76% accuracy range is clinically authentic for ILPD without synthetic data leakage (such as pre-split oversampling).

5. Interactive Streamlit Application & Cloud Deployment

- **Architecture:** Built with Streamlit, leveraging st.cache_resource for rapid in-memory pipeline inference and joblib model deserialization.
- **Features:** Dynamic algorithm selection dropdown, patient parameter validation sliders/number inputs, risk assessment gauges, and instant classification probability breakdowns.
- **Cloud Hosting:** Continuous deployment configured via GitHub repository integration on Streamlit Community Cloud at: **https://shreyash-hh-liver-disease-app-app-pcnfzj.streamlit.app/**

Student Declaration: *I hereby declare that this Formative Assessment (FA2) submission represents my authentic academic work and analysis completed for the Advanced Data Science Lab.*

Student Name: Shreyash Thakare | PRN: 125M1H048 | Date: 09-10-2026 | Status: Completed & Verified